This repository contains the input and solution data for the Train Formation and Shipment Path (TFSP) problem. The TFSP problem is critical to tactical railway freight planning, significantly impacting cost efficiency, asset utilization, and service reliability. The folder structure is illustrated in Figure 1.

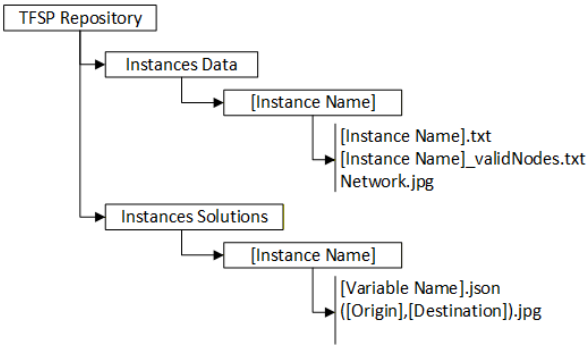

Figure 1: Folder Structure

The input data for instance P1_R_01_008 are as Table 1.

Table 1: Input files

File name	Description
P_R_01_008.txt	Shipment and network information of the instance
P_R_01_008_validNodes.txt	List of valid yards for transporting each shipment
Network.jpg	Image of the instance network

The solution data for instance P1_R_01_008 are as Table 2. In the problem solution, the values of the variables that take non-zero values are reported in the corresponding file.

Table 2: Solution files

Variable name	Description	File structure
X.json	This section indicates that shipment (i,j) traverses edge (k,t).	{ "Name": "X[i,j,k,t]", "Orig": "i", "Dest": "j", "Arc": { "O": "k", "D": "t" }, "val": 1 }
Y.json	This section indicates that shipment (i,j) is classified at yard k. If j = k, the shipment has been transported directly.	{ "Name": "Y[i,j,k]", "Orig": "i", "Dest": "j", "Class": "k", "val": 1 }
W.json	This section presents the volume of freight flow classified at yard k for shipments originating from node i and destined for node j.	{ "Name": "W[i,j,k]", "Orig": "i", "Dest": "j", "Class": "k", "val": 1 }

U.json	This section represents the total volume of blocks formed for the origin–destination (O–D) pair (i, j).	{
		"Name": "U[i,j]",
		"Orig": "i",
		"Dest": "j",
		"val": 1
		}

V.json	This section represents the number of sorting tracks used to form the block from origin i to destination j.	{
		"Name": "V[i,j]",
		"Orig": "i",
		"Dest": "j",
		"val": 1
		}

D.json	This section indicates that block (k,t) has departed from yard q together with train (i,j).	{
		"Name": "D[i,j,k,t,q]",
		"Orig": "i",
		"Dest": "j",
		"Block": { "O": "k", "D": "t" },
		"ExitNode": "q",
		"val": 1
		}

S.json	This section shows that the train service carrying block (k,t) has been switched at yard q.	{
		"Name": "C[k,t,q]",
		"Orig": "k",
		"Dest": "t",
		"ConnNode": "q",
		"val": 1
		}

(1,2).jpg To (8,7).jpg	Each image illustrates the transportation path of a shipment.	

Notes

- Standard parameters used across all instances are summarized in Table 3.

Table 3: Standard parameters

Parameter	Value
Fixed cost of train formation (FC)	400
Block swapping cost in intermediate yards (SC)	3
Maximum number of classification railcars per sorting track (a)	200
Detour ratio (γ)	1.35
Residual capacity rate (β_{st})	1
The proportional factor of the original capacity (θ_k)	1
Conversion factor (λ)	0.1

- The size of each train service is 50 wagons in all instances, except for instance P2_R_02_016, where it is considered to be 55 wagons.

An Illustrative Example

This example illustrates a railway network with 8 yards, labeled 1 through 8 in Figure 2. 9 bidirectional links are shown between the yards.

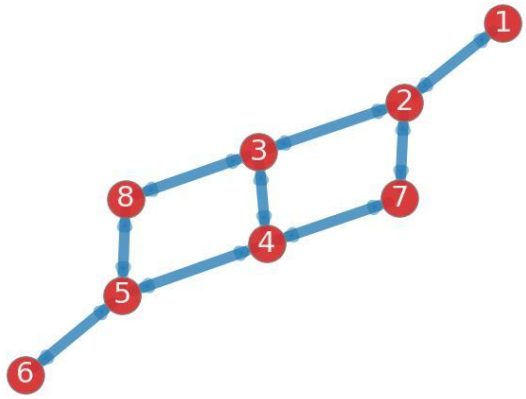

Figure 2: the rail network

Information regarding the network's capacities and its shipments is presented in the file P_R_01_008.txt with a structure similar to the one below.

n=8								Number of yards
m=18								Number of links
table_OD_pair								O-D pair matrix (volume of shipments matrix)
0	61	127	114	120	109	24	98	
98	0	102	124	85	85	21	47	
21	31	0	135	82	47	56	131	
104	82	53	0	72	42	41	136	
58	115	67	26	0	31	38	89	
132	127	39	135	74	0	43	71	
22	113	53	124	41	51	0	73	
48	127	64	42	137	137	39	0	
table_of_data_yard								The parameter values for yards. The columns, from left to right, represent the yard identifier, classification capacity(wagons), number of sorting tracks, classification cost, and accumulation cost.
1	64	6	4.2	12				
2	182	5	3.0	12				
3	492	6	3.0	12				
4	220	6	3.0	12				
5	265	5	4.0	12				
6	77	5	3.5	11				
7	261	6	4.2	11				
8	461	6	4.6	11				
table_of_data_link								The parameter values for links. The columns, from left to right, denote the link's origin, link's destination, link capacity, and its length. The link's capacity is expressed in terms of the number of trains.
1	2	14	200					
2	3	20	183					
3	4	15	225					
4	5	15	210					
5	6	14	340					
2	7	8	200					
7	4	8	220					
3	8	10	170					
8	5	10	240					
2	1	14	200					
3	2	20	183					
4	3	15	225					
5	4	15	210					
6	5	14	340					
7	2	8	200					
4	7	8	220					
8	3	10	170					

5	8	10	240					
BIG_M 4364								The total volume of all shipments.
shortest path								The shortest path length matrix between each origin and destination.
0	200	383	608	793	1133	400	553	
200	0	183	408	593	933	200	353	
383	183	0	225	410	750	383	170	
608	408	225	0	210	550	220	395	
793	593	410	210	0	340	430	240	
1133	933	750	550	340	0	770	580	
400	200	383	220	430	770	0	553	
553	353	170	395	240	580	553	0	

For example, shipment (7,6) is classified in yard 5. Its transportation path is depicted in Figure 3.

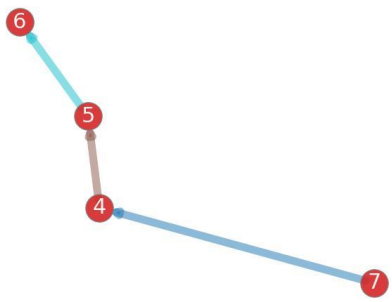

Figure 3: Shipment (7,6) transportation path

The variable values for this shipment are detailed in Table 4. This shipment was transported to yard 5 as block (7,5) in train service (7,5), and subsequently carried to its destination as block (5,6) in train service (2,6).

Table 4: Shipment (7,6) variable values

Variable Values		
{ "Name": "Y[7,6,5]", "Orig": "7", "Dest": "6", "Class": "5", "val": 1 }		
{ "Name": "W[7,6,5]", "Orig": "7", "Dest": "6", "Class": "5", "val": 51 }		
{ "Name": "X[7,6,7,4]", "Orig": "7", "Dest": "6", "Arc": { "O": "7", "D": "4" }, "val": 1 },	{ "Name": "X[7,6,4,5]", "Orig": "7", "Dest": "6", "Arc": { "O": "4", "D": "5" }, "val": 1 },	{ "Name": "X[7,6,5,6]", "Orig": "7", "Dest": "6", "Arc": { "O": "5", "D": "6" }, "val": 1 }

```
{
  "Name": "D[7,5,7,5,7]",
  "Orig": "7",
  "Dest": "5",
  "Block": {"O": "7", "D": "5"},
  "ExitNode": "7",
  "val": 1
},
{
  "Name": "D[7,5,7,5,4]",
  "Orig": "7",
  "Dest": "5",
  "Block": {"O": "7", "D": "5"},
  "ExitNode": "4",
  "val": 1
},
{
  "Name": "D[2,6,5,6,5]",
  "Orig": "2",
  "Dest": "6",
  "Block": { "O": "5", "D": "6" },
  "ExitNode": "5",
  "val": 1
}
```